



1. How Good People Can Draw?

[QuickDraw](#) is an online game developed by Google in which the players are challenged to draw a picture of an object, then attempts to guess what the drawings represent using a neural network artificial intelligence. The AI is capable of learning from each drawing and therefore increasing its accuracy in the future.

Using the drawings made by millions of players, a dataset with 50 million drawings from 345 objects has been created. Here, a subset of this gigantic dataset, [QuickDraw10](#), is going to be used for an image clustering task.



Figure. In QuickDraw, player has to draw the given objects as much accurate as

- Perform PCA on the dataset to reduce its dimensions. Display the top 20 eigenvalues as well as the representation of the samples projected onto their first two and three principal components.
- Now use LDA to project the samples onto their first and second linear discriminants, and display the results.
- Perform K-Means on the data using their first two principal components. Select random initial centroids and set $K = 3, 7, 10$. Display the result of clustering with final centres highlighted, and compare the results together and with the plot obtained in part (a).
- Repeat part (c) considering the initial centroids as the mean of the samples such that for $K = 3$, initial centroids are the mean of samples of classes $\{1,3,5,7\}$, $\{2,4\}$, and $\{6,8,9,10\}$, for $K = 7$, initial centroids are the mean of samples of classes $\{1,2,5\}$, $\{3\}$, $\{4\}$, $\{6\}$, $\{7\}$, $\{9\}$ and $\{8,10\}$, and for $K = 10$, initial centroids are the mean of sample of each of the 10 classes separately. Display and compare the results with those of the previous parts. Also comment on your observations.
- As can be seen, clustering the data using their first two principal components doesn't produce satisfactory results. We are going to increase the clustering accuracy by considering more principal components. By trial and error find the number of principal components which is enough to capture 0.95 of the data variance. Reconstruct three arbitrary samples using these principal components and compare the results with their corresponding original images.

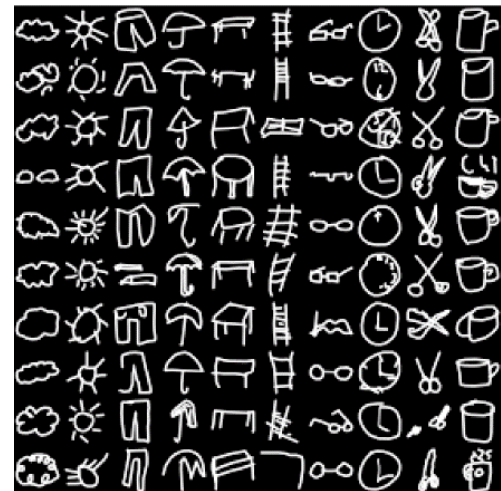


Figure. QuickDraw10 is a subset of QuickDraw dataset, a commonly

- f. Use K-Means to divide data into 10 clusters. Set the required parameters appropriately. Display 10 samples of each cluster randomly (100 in total) and comment on the results.
- g. Draw 10 bar graphs for each cluster, each representing the distribution of the samples of different classes (in percentage). Compare different clusters using these bar graphs.
- h. Finally, visualize the clustering by keeping only 2 and 3 features and displaying the corresponding scatter plots.

2. Rainy or Sunny? K-NN Answers

The *Bureau of Meteorology*, or *BOM*, is an Australian governmental organization which is responsible for monitoring weather statistics and providing forecasts, warnings, and observations to the Australian public. With over seven thousand weather stations across the country, BOM records a variety of aspects of the weather in different regions, such as temperature, humidity, rainfall, air pressure, sunshine, wind speed and direction. A dataset of daily climate records gathered from different BOM stations between 2008 and 2017 is given, and the objective is to determine whether tomorrow is going to be rainy or not. First, remove those samples in which the target attribute has unassigned value. As for the remaining attributes with null values, you must set them to the median of the values of that attribute across the same exact month. To accomplish this, you can define a new column 'month' which contains the corresponding month according to the value of 'date'. Please note that you can also use the values of this column as a feature. Finally, partition the dataset into three sets 'train', 'test', and 'validation' according to appropriate proportions.

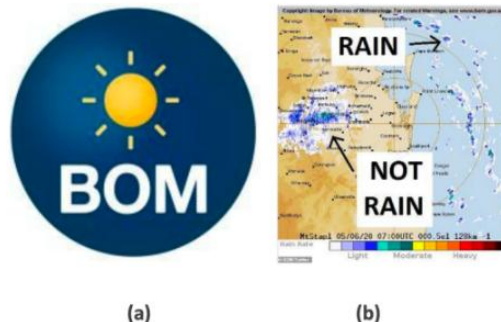


Figure 3 For more than a century, BOM has been responsible for measuring and publishing weather statistics. (a) The organization logo. (b) A live map posted on BOM Twitter account.

- a. To prevent overfitting, we first aim to reduce the number of variables. One possible method is to detect highly correlated features and eliminate either of them from the dataset. Find the feature pairs, if any, with correlation coefficient over 0.95. Is it reasonable to keep these features in the dataset? If so, remove one of the features of each pair to your liking.

- b. Next, determine the 10 best features of the dataset using sequential feature selection method.
- c. Apply K-NN to the modified dataset. Find the best value of k using the validation set, and report the accuracy of the model on the test set.
- d. Use the same model to forecast weather condition in the city of Rasht based on the city's similar records. You can treat the missing values similar to the previous part. Justify your observations.
- e. Is it possible to determine the probability of rainfall using the same strategy? If so, how?

Note: There is no restriction on the usage of built-in functions and libraries

3. Predicting Health Outcomes: Can We Accurately Diagnose Diseases Using Machine Learning?

In the modern world, machine learning models are increasingly used to assist in diagnosing health conditions based on patient data. In this assignment, you will explore two different real-world datasets related to medical diagnoses and apply machine learning techniques to classify patient outcomes.

Depending on the last digit of your student number, you will work with one of the following datasets:

- Odd digit: Diabetes dataset
https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_diabetes.html
- Even digit: Breast Cancer dataset
https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html

Your task is to preprocess, explore, and model the data using a variety of techniques.

a. Data Preprocessing and EDA:

- Load the assigned dataset.
- Perform data cleaning if necessary.
- Conduct Exploratory Data Analysis (EDA) to understand the distribution of features, detect missing values, and observe relationships between features and the target variable.
- Visualize important aspects of the data (for example: feature distributions, correlations, and class distributions).

b. Model Benchmarking with Lazy Predict:

- Use the Lazy Predict library to automatically compare a variety of classification models on your dataset.
- Report the accuracy, precision, recall, and F1-score for each model produced by Lazy Predict.
- Identify the best-performing model based on F1-score.

c. Focused Modeling with Scikit-Learn:

- Take the best model identified from Lazy Predict results and manually implement it using Scikit-Learn (do not use Lazy Predict for this step).
- Perform hyperparameter tuning using GridSearchCV or RandomizedSearchCV.
- Report the tuned parameters and the resulting model's performance.

d. Evaluation and Reporting:

- Compare the model's performance before and after hyperparameter tuning.
- Plot a confusion matrix for the final tuned model.
- Discuss the advantages and limitations of using automated model selection tools like Lazy Predict in medical data analysis.
- Identify which features contribute the most to the classification task according to your final model

4. Optical Character Recognition Through Minimum Distance Classifier.

A Minimum Distance Classifier attempts to classify an unlabelled sample to a class which minimizes the distance between the sample and the class in multi-feature space. As minimizing distance is a measure for maximizing similarity, MDC actually assigns data to its most similar category.

While MDC might look too basic, it works pretty well in some problems. One of them could be Optical Character Recognition (OCR), where the goal is to distinguish handwritten or printed text characters inside digital images of documents. You are working with a customised dataset here, which is available in the 'P2' folder of the 'inputs' directory. The dataset is divided into three groups, as depicted in Figure 2.

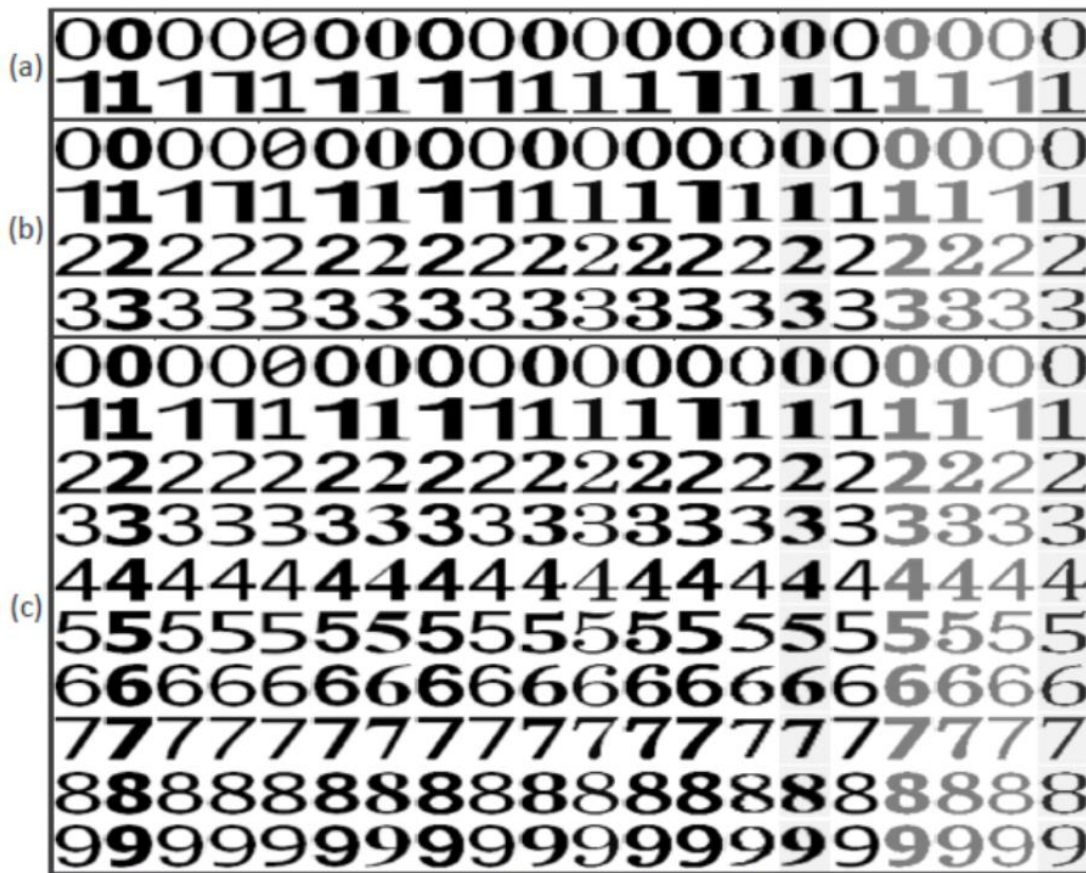


Figure 2 The problem is divided into three different parts with three different datasets (a) first dataset, including only two digits with structurally distinctive appearance (b) second dataset, which leads to a more complicated task with four categories (c) last dataset, including all digits.

First, consider group (a).

- Using the train images, find and display the prototype of each of the available classes.
- Use test samples to evaluate your MDC classifier. Report the error.
- Repeat part a. and b. with group (b).
- Repeat part a. and b. with group (c).
- Comment on the results obtained from the previous parts. What were your observations?

Good Luck!
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